QSL
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QuantWorld

Quantum Hackathon

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Finance

What makes an investment decision?

An investment decision connects:

Purpose

What should the money achieve?

Horizon

When will it be needed?

Exposure

What outcomes could affect the investor?

Restrictions

Which decisions are acceptable?

Opening question:

Should two investors with the same market information choose the same portfolio?

Markowitz (1952); Boyd et al. (2017).

Different tasks within the investment process

Task	Question
Define the decision	What are the objectives and restrictions?
Model possible outcomes	What could happen, and under which assumptions?
Estimate inputs	What can available evidence tell us?
Choose an allocation	Which holdings or allocation rule should we use?
Evaluate and monitor	What would justify acting or reconsidering?

Takeaway: Estimating outcomes, choosing holdings, and evaluating a recommendation are distinct tasks.

Boyd et al. (2017).

Return: the change relative to the amount invested

Return measures the gain or loss over a specified period relative to the starting investment.

For asset i, during period t:

- $r_{i,t}$ realised return.
- $P_{i,t-1}$ beginning price.
- $P_{i,t}$ ending price.
- $D_{i,t}$ cash distribution per unit held.

$$r_{i,t} = \frac{P_{i,t} - P_{i,t-1} + D_{i,t}}{P_{i,t-1}}.$$

Example

Beginning price €100, ending price €104, distribution €2:

$$r = \frac{104 - 100 + 2}{100} = 6\%.$$

French (n.d.), Data Library; Stopfer and Wagner (2026), Appendix A. Example: original teaching calculation.

A portfolio is an allocation

A portfolio is a collection of investment positions.

For one period:

- n number of assets.
- w_i beginning-of-period fraction invested in asset i.
- r_i realised return of asset i.
- r_p realised portfolio return.

Example $r_p = 0.60(8\%) + 0.40(-2\%) = 4\%.$

For a fully invested portfolio without short positions:

$$\sum_{i=1}^{n} w_i = 1, \qquad w_i \geq 0.$$

Before costs:

$$r_p = \sum_{i=1}^{n} w_i r_i.$$

Markowitz (1952); Sakuler et al. (2025), Section 2.

Observed, uncertain, and estimated returns

Use different notation for different concepts:

Symbol	Meaning
r_i	An observed return
R_i	An uncertain future return represented by a random variable
μ_i	Expected return under the model
$\hat{\mu}_i$	An estimate of expected return

Here $\mathbb{E}$ means the probability-weighted average:

$$\mu_i = \mathbb{E}[R_i].$$

For fixed portfolio weights:

$$R_p = \sum_i w_i R_i, \qquad \mu_p = \mathbb{E}[R_p] = \sum_i w_i \mu_i.$$

Takeaway: An expected return is a model quantity, not a promised outcome.

Markowitz (1952); Sharpe (1994), historical and prospective quantities.

Risk begins with possible consequences

To describe investment risk, specify:

- The exposure and time horizon.
- What counts as an adverse outcome.
- Its possible magnitude and likelihood.
- The assumptions behind that description.

For a simple portfolio-loss model:

R_p	uncertain portfolio return.
$L = -R_p$	loss as a fraction of starting wealth.
ℓ	a loss threshold.
$\Pr$	probability under the model.

$$F_L(\ell) = \Pr(L \le \ell).$$

Takeaway: A loss distribution describes a profile of possible outcomes; one number summarises only part of it.

Artzner et al. (1999); Rockafellar and Uryasev (2002).

Likelihood and severity: expected loss

Let:

L	loss, with gains represented by negative values.
$L_+ = \max(L, 0)$	the positive loss amount.
$\mathbb{E}[L \mid L > 0]$	average loss conditional on a loss occurring.

When a loss has positive probability:

$$\mathbb{E}[L_+] = \Pr(L > 0)\,\mathbb{E}[L \mid L > 0].$$

Same expected loss, different profiles

Illustration	Probability of loss	Loss when it occurs	Expected positive loss
A	100%	1%	1%
B	1%	100%	1%

Takeaway: An average loss does not reveal how losses are distributed.

Shevchenko (2010), frequency–severity loss modelling. Identity and examples: original teaching derivations.

Volatility is one measure of dispersion

For uncertain return R_i with mean μ_i:

σ_i^2 return variance.

σ_i return standard deviation, or volatility.

$$\sigma_i^2 = \mathbb{E}[(R_i - \mu_i)^2], \qquad \sigma_i = \sqrt{\sigma_i^2}.$$

Variance measures deviations above and below the mean.

Measure	What it summarises
Volatility	Dispersion of returns
Loss probability	Likelihood of a specified adverse outcome
Expected positive loss	Average positive loss
Expected shortfall	Average loss in the worst specified fraction of outcomes

Takeaway: The appropriate risk measure depends on the decision.

Markowitz (1952); Rockafellar and Uryasev (2002).

Dependence: how assets behave together

For returns R_i, R_j and means μ_i, μ_j:

Σ_{ij} covariance.

ρ_{ij} correlation.

$$\Sigma_{ij} = \mathbb{E}[(R_i - \mu_i)(R_j - \mu_j)].$$

If both volatilities are positive:

$$\rho_{ij} = \frac{\Sigma_{ij}}{\sigma_i \sigma_j}.$$

- Positive correlation: positive linear association.
- Negative correlation: negative linear association.
- Zero correlation: no linear association.

Takeaway: Individual asset characteristics do not fully describe portfolio outcomes.

Markowitz (1952); Sharpe (1994).

Diversification and portfolio variance

For two assets A and B:

$$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B \rho_{AB}\sigma_A\sigma_B.$$

Here σ_p^2 is portfolio-return variance.

Illustration: equal weights; each asset has 20% volatility.

Correlation	Portfolio volatility
1	20.0%
0	14.1%
−0.5	10.0%

Takeaway: Combining assets can reduce variance when their returns are less than perfectly positively correlated.

Markowitz (1952). Numerical example: original calculation.

Objectives, risk criteria, and constraints

An illustrative mean–variance allocation model uses:

w	vector of portfolio weights.
$\hat{\mu}$	vector of estimated expected returns.
$\hat{\Sigma}$	estimated covariance matrix.
$\mathcal{W}$	set of permitted allocations.
$\lambda > 0$	chosen weight on variance.

The superscript $^\top$ denotes transpose.

$$\max_{w \in \mathcal{W}} \left[w^\top \hat{\mu} - \frac{\lambda}{2} w^\top \hat{\Sigma} w \right].$$

In words:

Choose a permitted allocation that maximises estimated estimated return minus a weighted variance term.

Boyd et al. (2017); Rockafellar and Uryasev (2002), alternative risk criteria.

Estimation choices enter the decision

Let $T > 1$ be the number of observed periods and $r_{i,t}$ the observed returns.

A historical sample mean:

$$\hat{\mu}_i = \frac{1}{T} \sum_{t=1}^{T} r_{i,t}.$$

A sample covariance:

$$\hat{\Sigma}_{ij} = \frac{1}{T-1} \sum_{t=1}^{T} (r_{i,t} - \hat{\mu}_i)(r_{j,t} - \hat{\mu}_j).$$

Choices to document

- Observation window
- Frequency
- Asset universe
- Data treatment
- Estimator

Ledoit and Wolf (2004); French (n.d.).

Sensitivity: what would change the recommendation?

Let $\Delta\mu$ and $\Delta\Sigma$ be changes in assumed means and covariances.

For fixed weights:

$$\text{Change in expected return} = w^\top \Delta\mu$$

$$\text{Change in variance} = w^\top \Delta\Sigma\, w.$$

Then investigate:

- Does choosing the allocation again change the holdings?
- Which assumption explains the change?
- What remains stable?

Takeaway: Stability is conditional on the assumptions tested.

Ledoit and Wolf (2004); DeMiguel et al. (2009). Equations: direct derivations.

Rebalancing: improvement relative to what?

w^-	current weights immediately before trading.
w	proposed weights.
$\Delta w = w - w^-$	proposed allocation change.
$C(\Delta w)$	estimated trading cost as a fraction of wealth.

An illustrative score:

$$J(w) = w^\top \hat{\mu} - \frac{\lambda}{2} w^\top \hat{\Sigma} w - C(\Delta w).$$

Question:

Does the proposed change justify its costs relative to keeping the current holdings?

Boyd et al. (2017); Mugel et al. (2022).

Investigations without a new allocation optimum

Approach	Possible investigation
Equal weighting	What does a transparent allocation rule achieve?
Retaining holdings	When is avoiding a trade defensible?
Prescribed target	What happens when weights drift?
Threshold rebalancing	How do thresholds affect trading and outcomes?
Scenario evaluation	How does an existing portfolio behave under changed assumptions?

For n assets, equal weighting sets:

$$w_i = \frac{1}{n}.$$

DeMiguel et al. (2009). Investigation menu: teaching synthesis.

Where quantum methods may enter

Discrete allocation

$x_i = 1$ select asset i;

$x_i = 0$ do not select it.

K a fixed positive number of selected assets.

For equal investment among selected assets:

$$\sum_i x_i = K, \qquad w_i = \frac{x_i}{K}.$$

With mean vector μ and covariance matrix Σ:

$$\mu_p = \frac{\mu^\top x}{K}, \qquad \sigma_p^2 = \frac{x^\top \Sigma x}{K^2}.$$

Other tasks: estimate expected payoffs under a specified probability model.

Sakuler et al. (2025); Rebentrost et al. (2018). Selection equations: teaching derivation.

Verify the output and identify the resources

Approach	What must be established?
Exact method	Feasibility and any claimed optimality certificate
Classical heuristic	Candidate quality and repeatability
Quantum-inspired method	What makes the method quantum-inspired?
Hybrid quantum method	Which steps are classical and quantum?
Quantum sampling	Which sampled decisions are feasible and useful?

Takeaway: The financial meaning of an output depends on its formulation and verification.

Farhi et al. (2014); Mugel et al. (2022); Stopfer and Wagner (2026).

Possible directions for further investigation

Your main contribution could concern:

- Investment objectives and defensible trade-offs.
- Estimation and sensitivity.
- Retention or rebalancing rules.
- Financial meaning of computational formulations.
- Evidence needed before acting.

Distinguish uncertainty about:

Future outcomes | Model and parameter estimates | Computational results

Ledoit and Wolf (2004); DeMiguel et al. (2009); Sakuler et al. (2025). Directions: teaching synthesis.



Hands on

Team exercise: What does the evidence establish?

Using the assigned excerpt and results:

1. Decision
What financial question is being investigated?

2. Model
What objectives, inputs, and restrictions are used?

3. Method
What does the computational approach contribute?

4. Evidence
What is supported, and under which conditions?

5. Next step
What remains worth investigating?

Output: one slide containing:

- Question
- Formulation
- Finding
- Limitation
- Next question

The three assigned primary papers.

What would justify using the result?

Each team reports:

> “The study supports ___ under ___. Before using this finding for ___, we would need ___.”

Discuss

- Does the result establish mathematical solution quality, financial performance, or both?
- Which restrictions must hold?
- Which assumptions could reverse the conclusion?
- What evidence supports acting, retaining holdings, or investigating further?

Sharpe (1994); assigned primary papers.

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